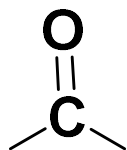
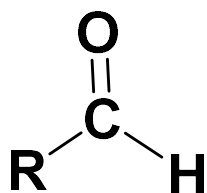


UNIT 6

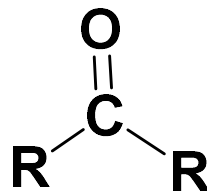
CARBONYL REACTIONS



Carbonyl



Aldehyde



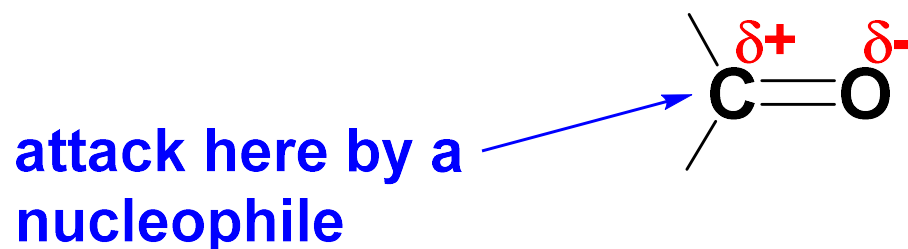
Ketone

Carbonyl Reaction

- Addition
- Addition – Elimination
- Enolization-ketolization

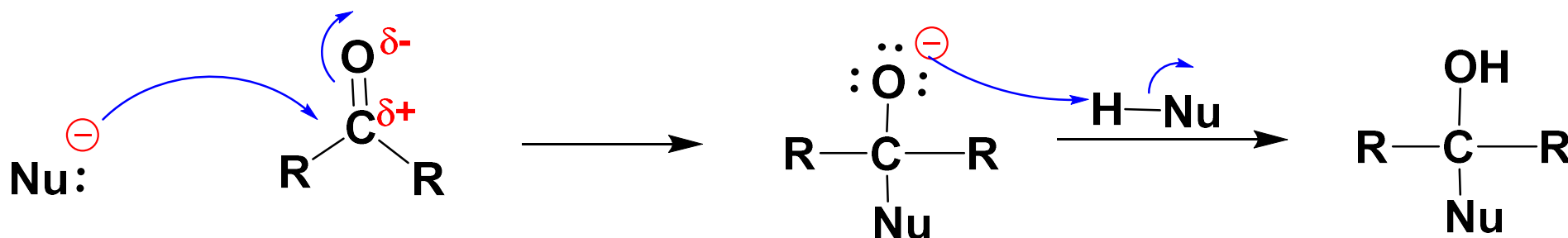
6.1. Carbonyl addition

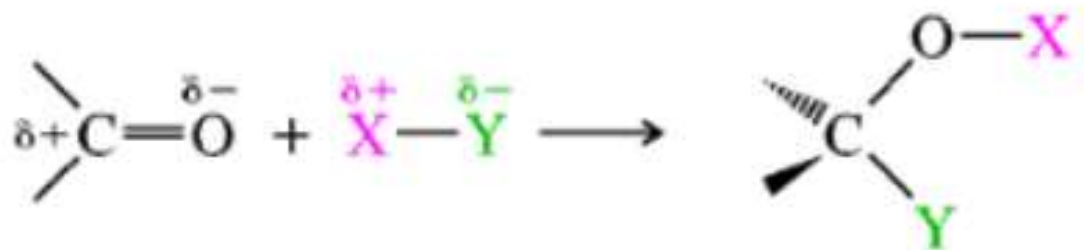
Oxygen is more electronegative than carbon. Therefore the electrons in the C=O bond are attracted to oxygen, producing a highly polarized bond.



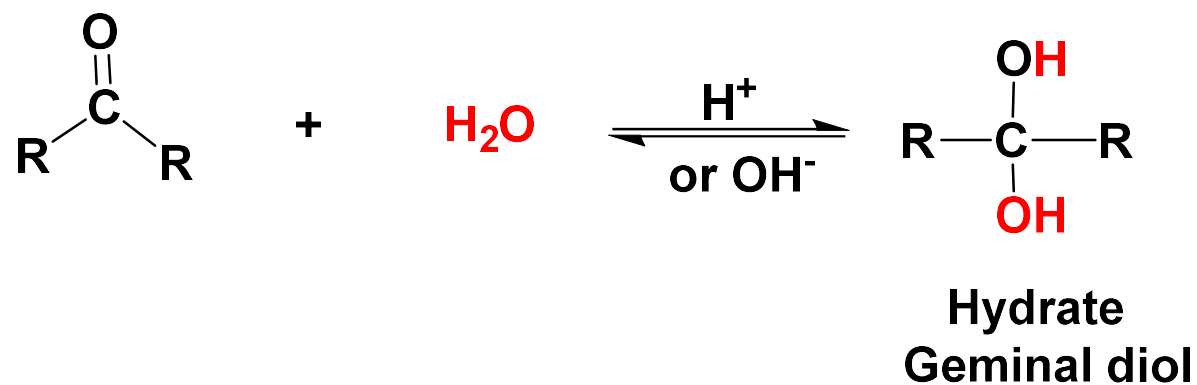
Oxygen is more electronegative than carbon. Therefore the electrons in the C=O bond are attracted to oxygen, producing a highly polarized bond.

- Nucleophiles can add to carbonyl group under either acidic or basic condition

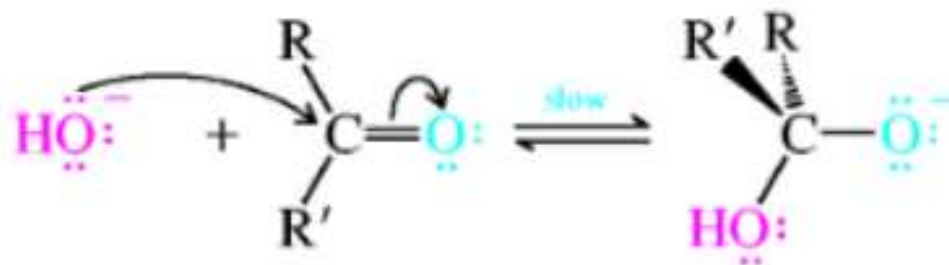




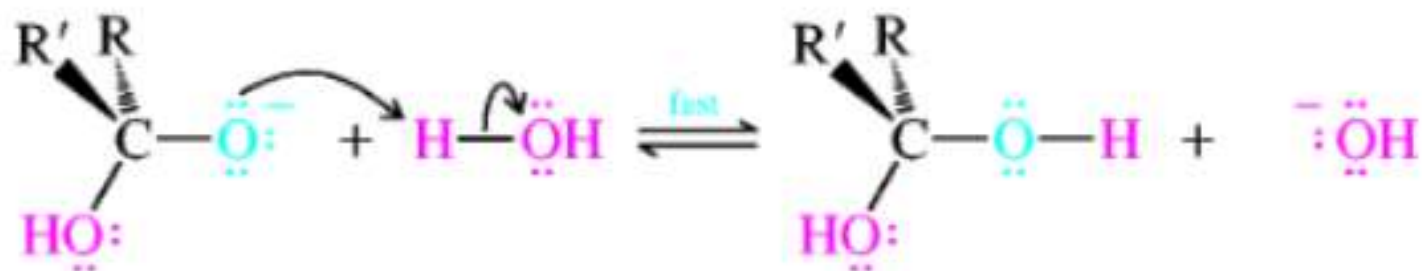
1. Hydration – Addition of H₂O



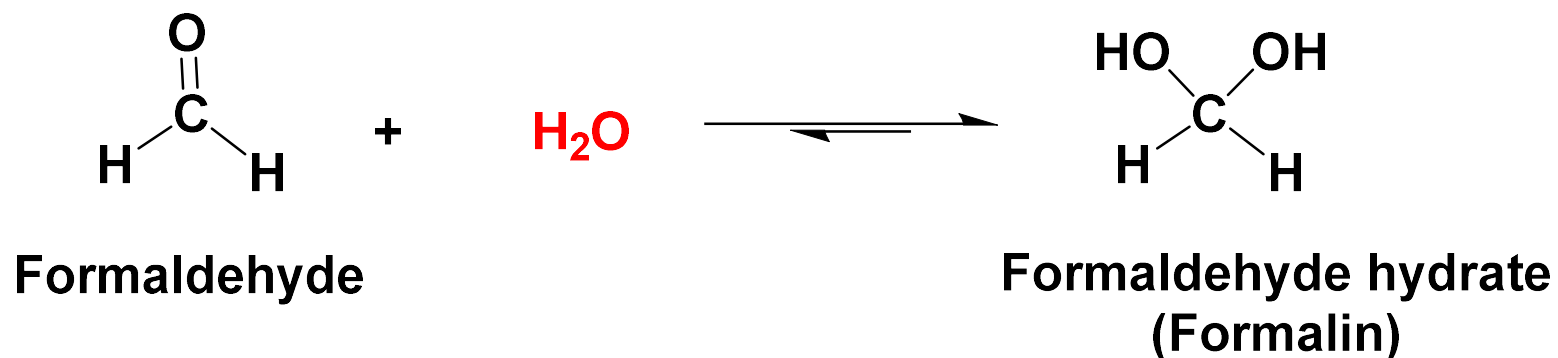
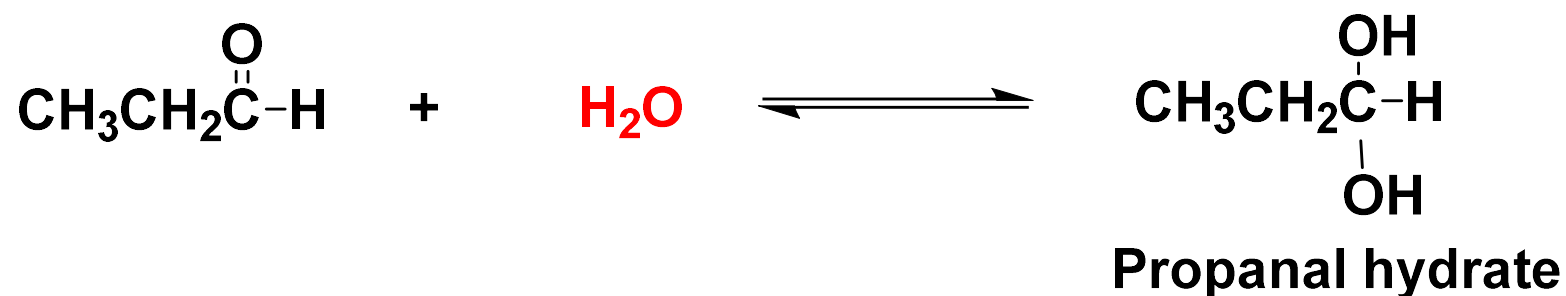
Step 1: Nucleophilic addition of hydroxide ion to the carbonyl group.



Step 2: Proton transfer from water to the intermediate formed in step 1.

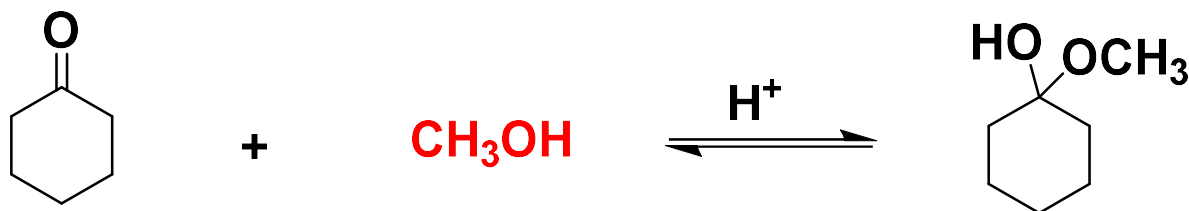
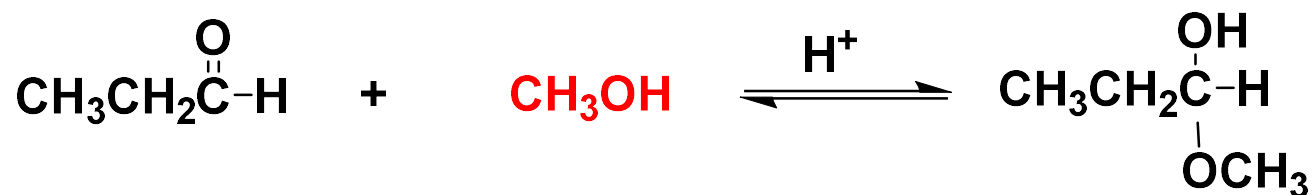
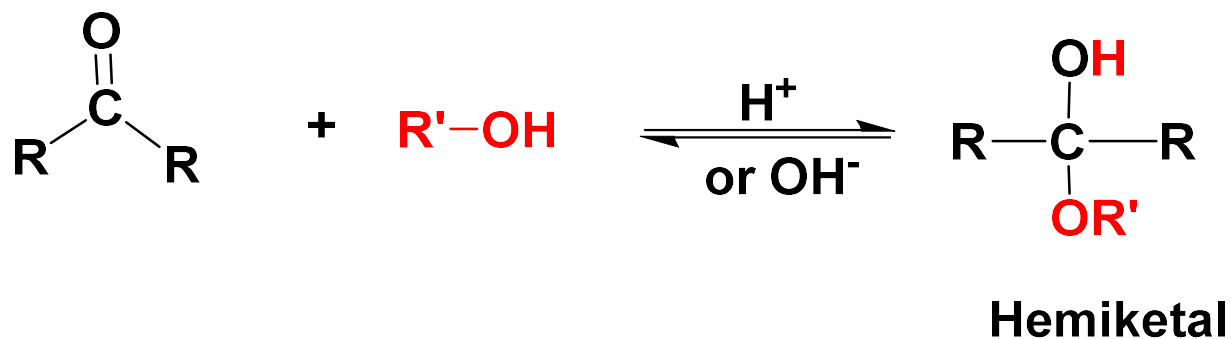
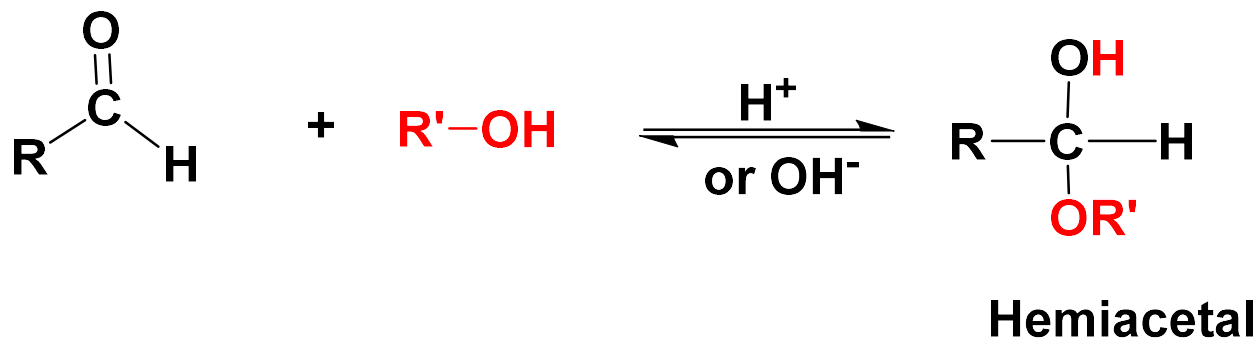


- Hydrates are unstable species. Backward reaction may occur.

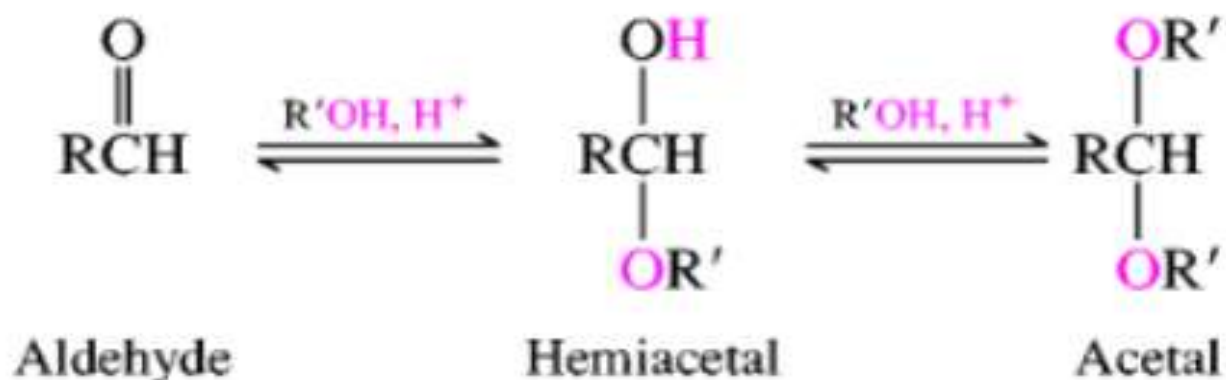


2. Acetal Formation

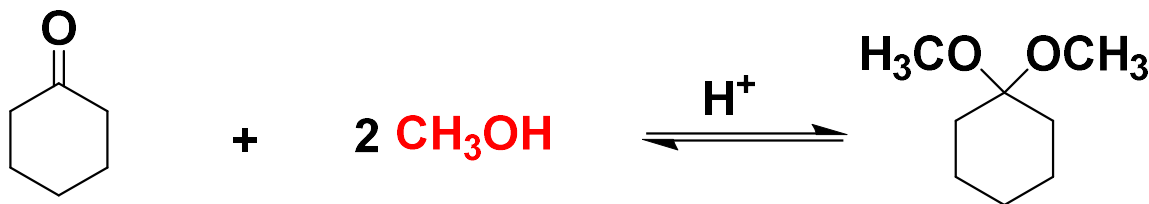
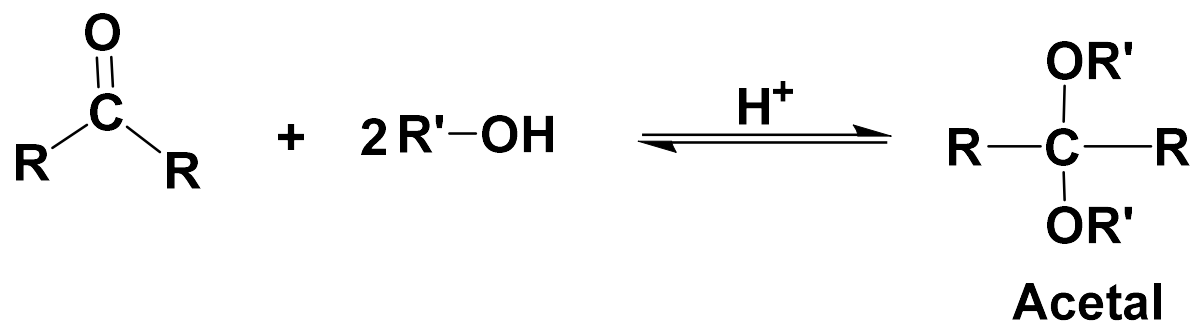
- Aldehydes and ketones react with alcohols to form acetals .



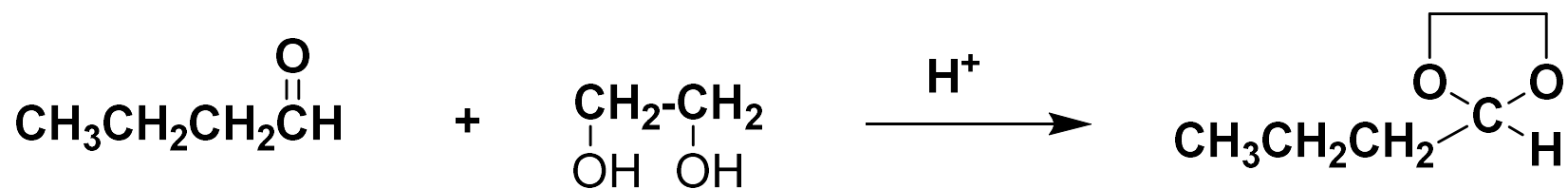
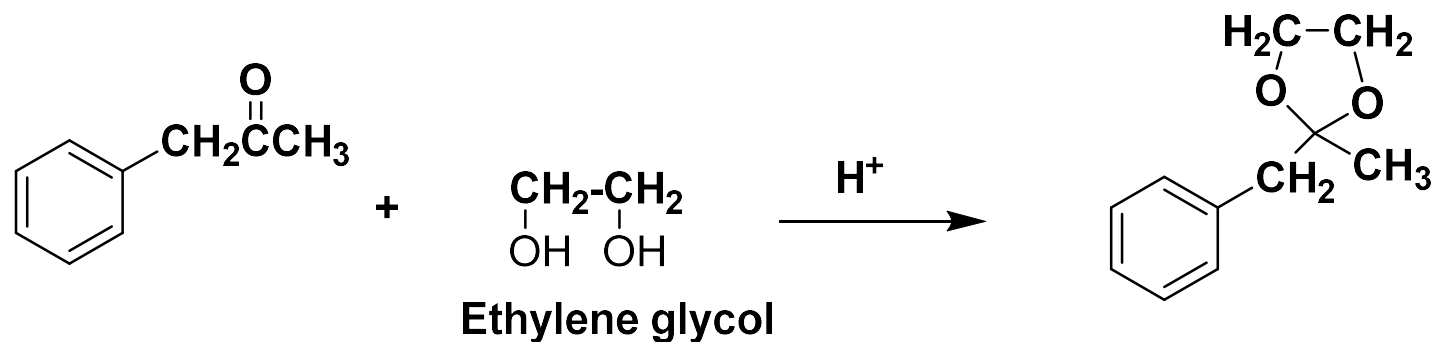
- Hemiacetals and ketals are not stable, can change to acetals.



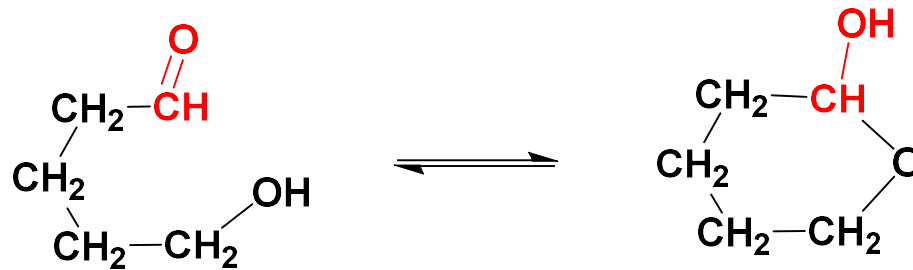
- In the formation of acetals two molecules of alcohols adds to the carbonyl.



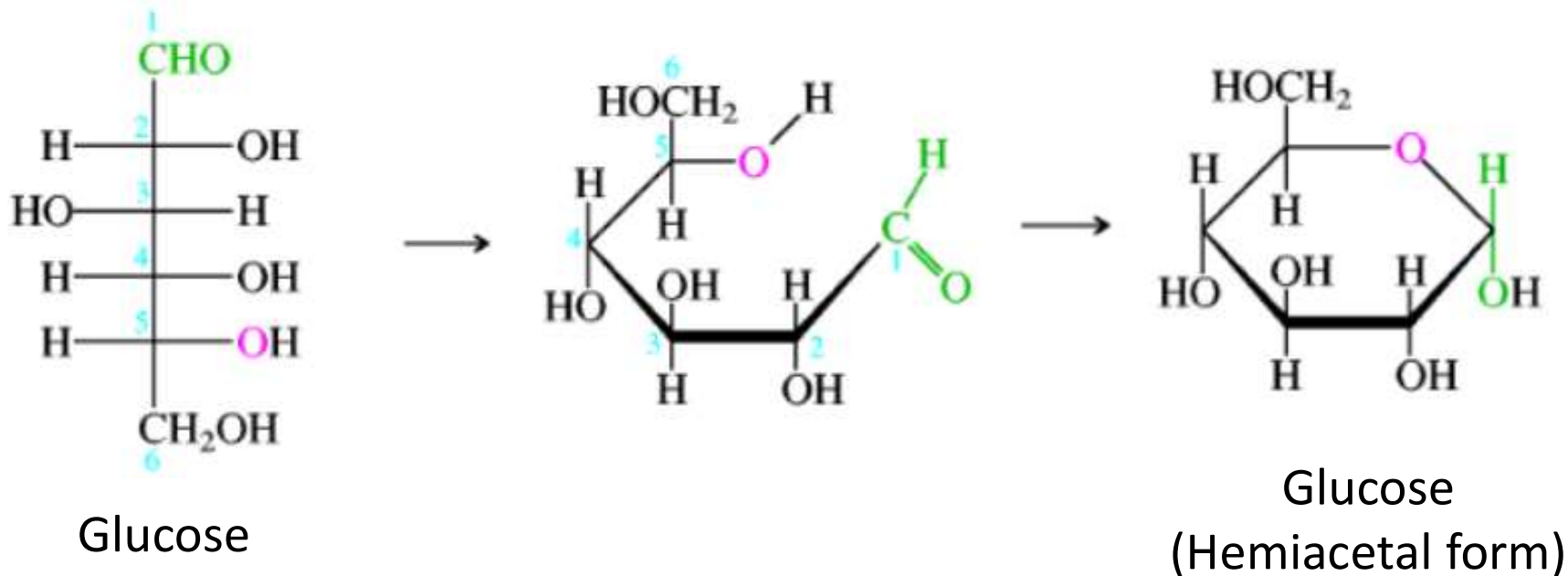
- Diols that bear two hydroxyl groups in a 1,2 or 1,3 relationship to each other yield cyclic acetals on reaction with either aldehydes or ketones.



- When the carbonyl and the –OH groups found in the same molecule, cyclic hemiacetals formed.

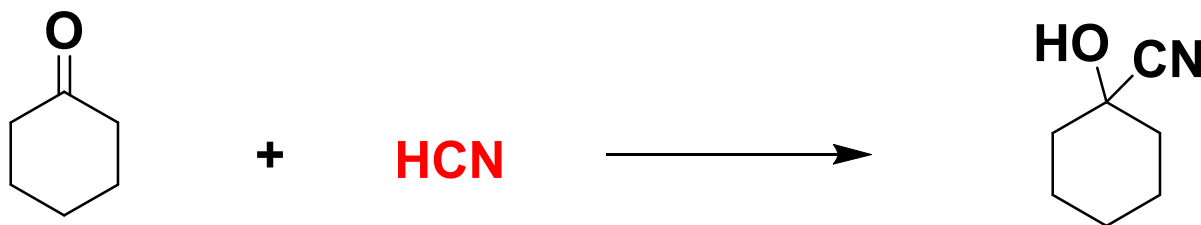
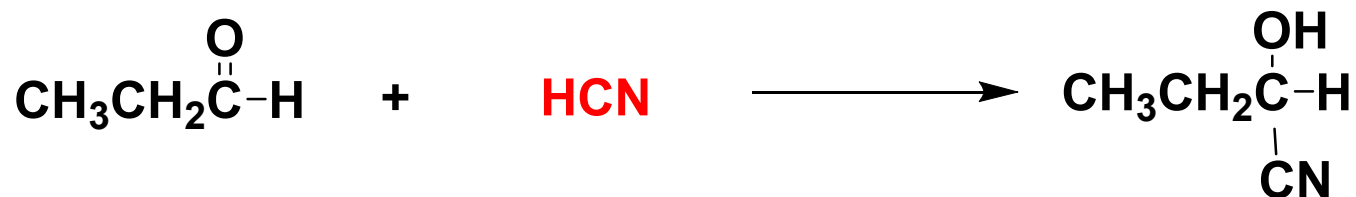
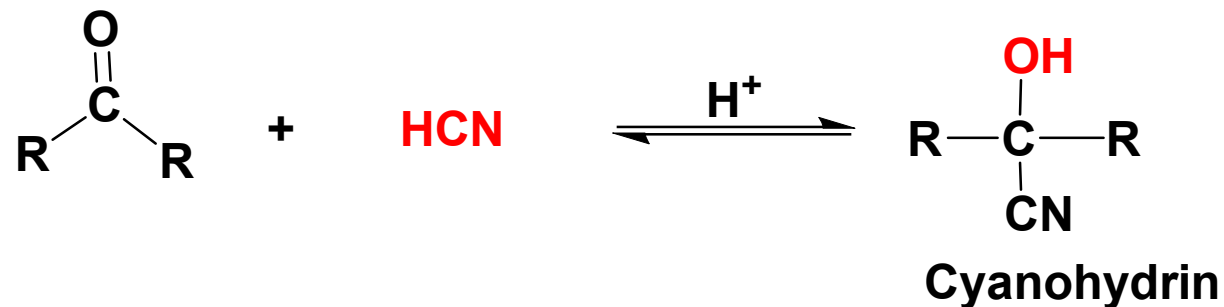


- Glucose exist as a cyclic hemiacetal.



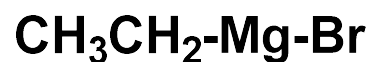
3. Addition of Hydrogen cyanide: Cyanohydrins

If HCN is added to carbonyl, a cyanohydrin results.

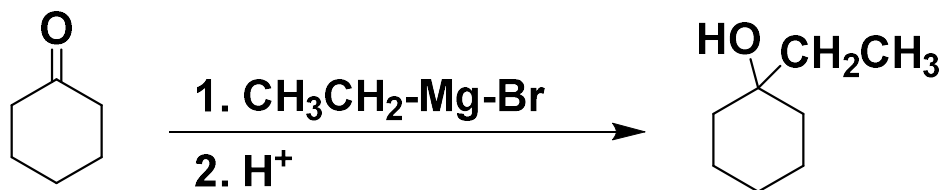
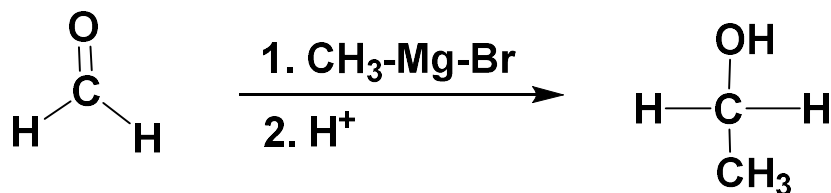
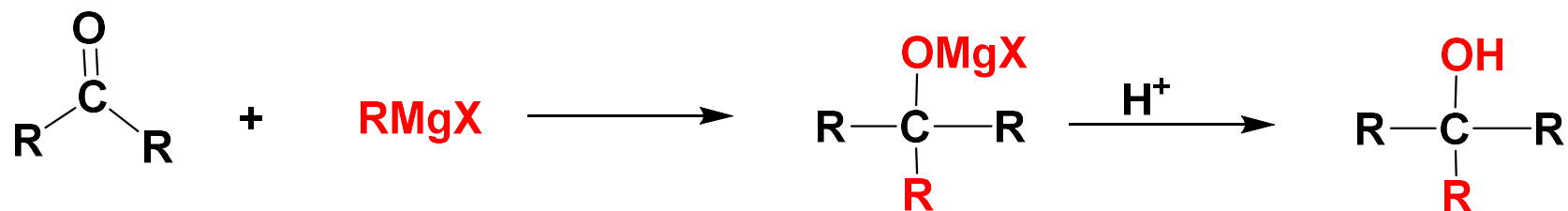


3. Addition of Grignard Reagents

Grignard reagent-organometallic compound

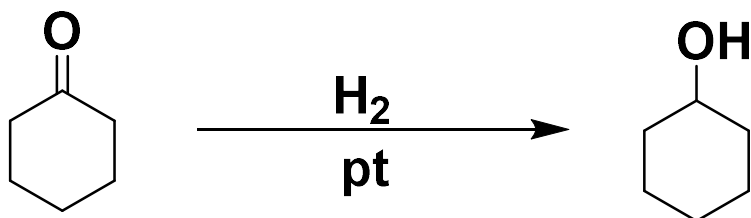
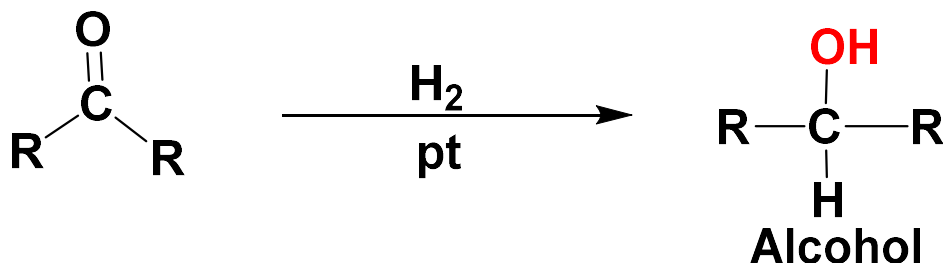


Grignard reagent adds to carbonyl to form alcohols



4. Addition of Hydrogen

Catalysed by Pt, Pd, or Ni



4. Reduction of Carbonyl

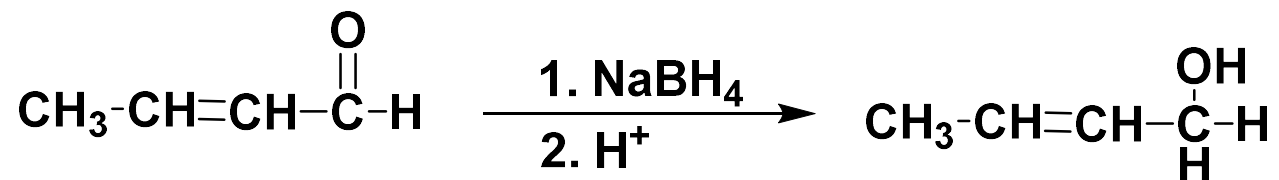
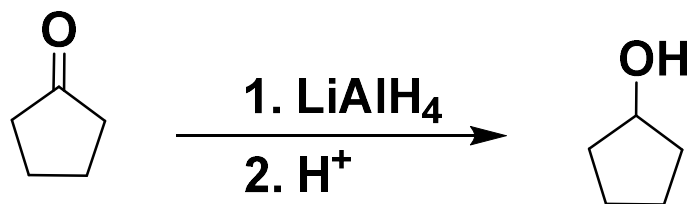
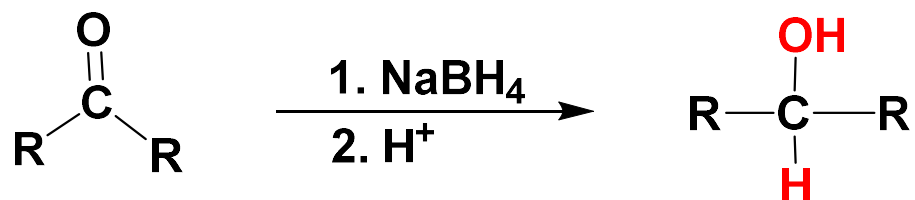
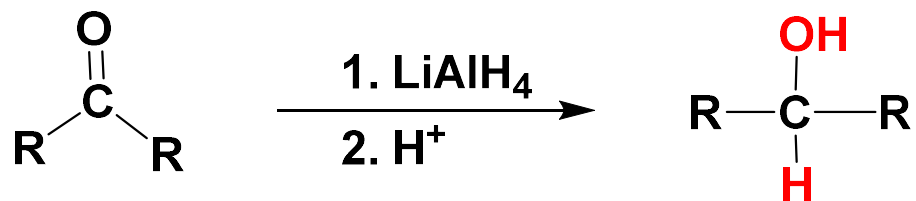
The most common metal hydride used to reduce carbonyl compounds are Lithium aluminium hydride and sodium borohydride



Lithium aluminium hydride



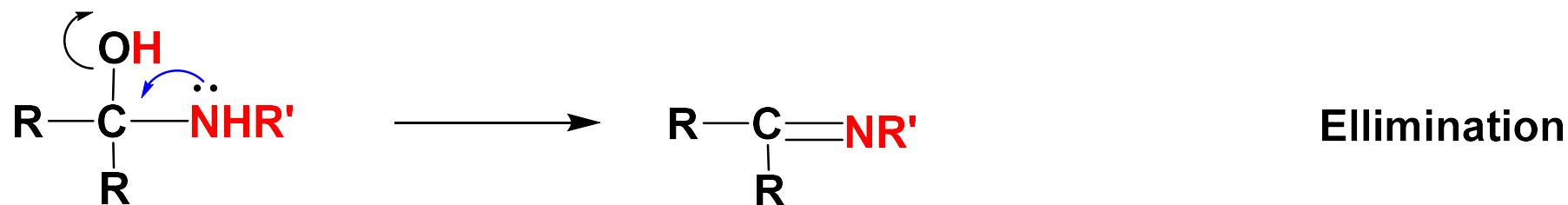
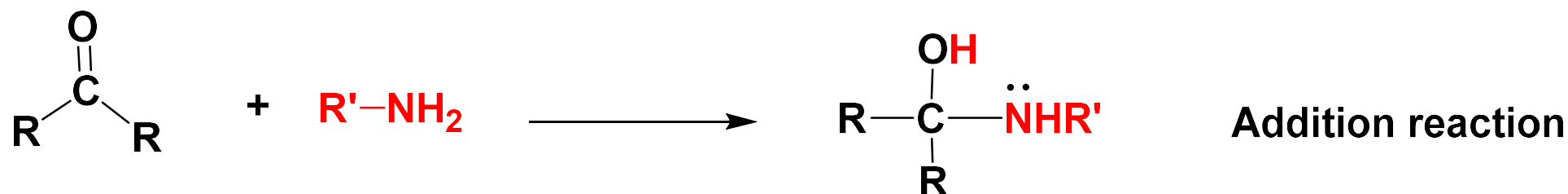
Sodium borohydride



Reduce only the carbonyl not the C-C double bond

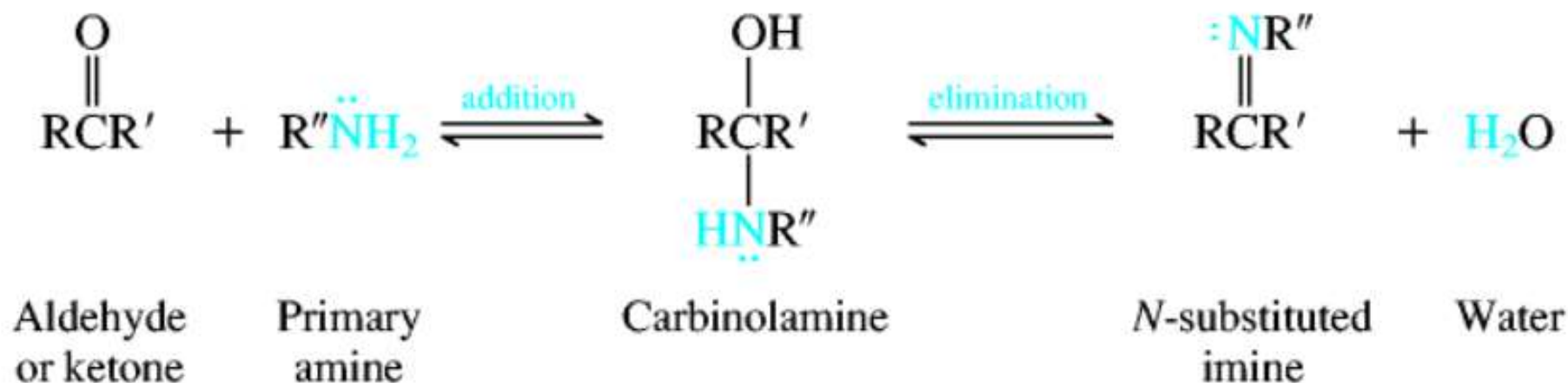
6.2. Addition – Elimination reaction

If the added Nu contain unshared electrons, it may donate them to form a double bond, thereby eliminating the original carbonyl oxygen atom.

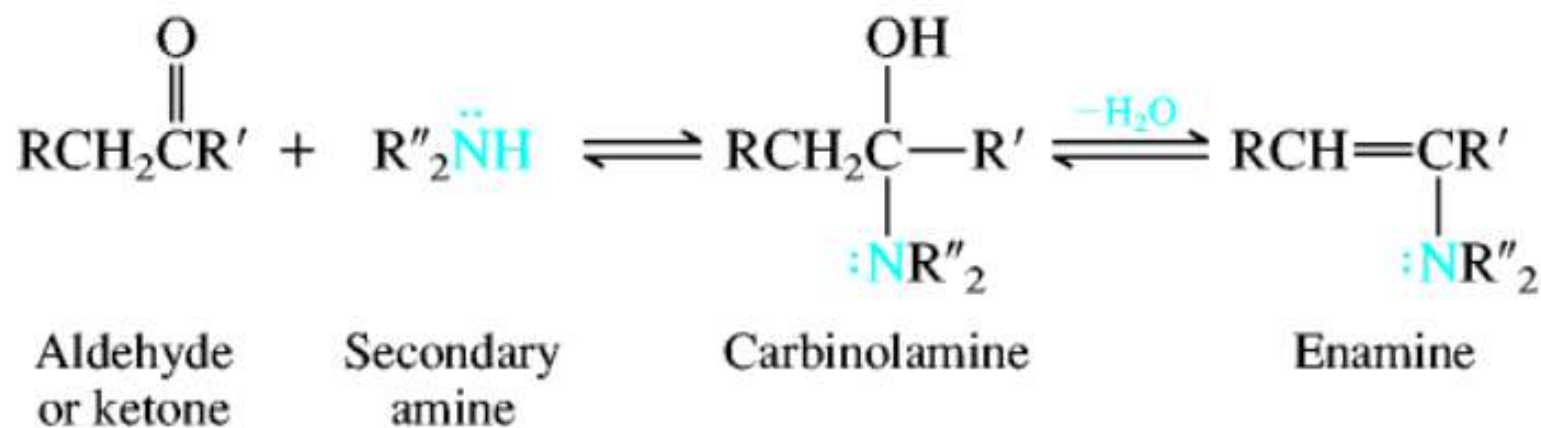


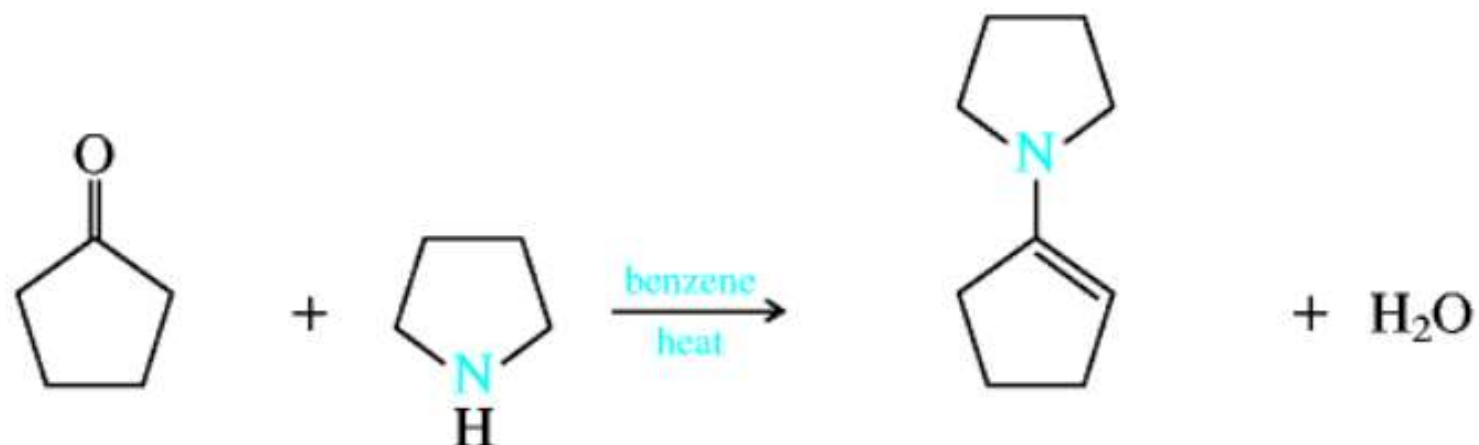
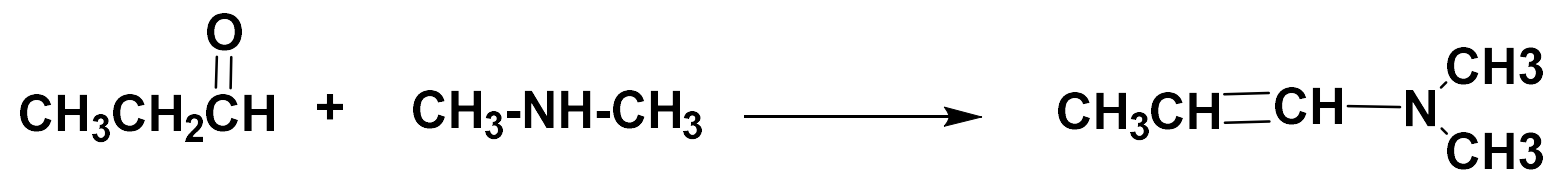
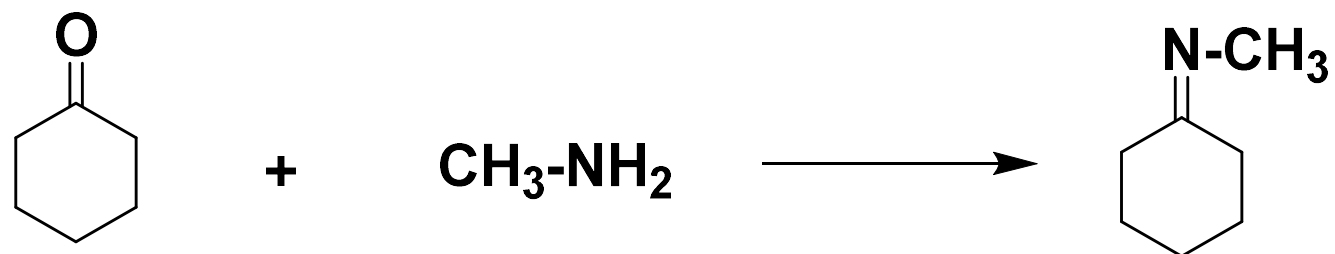
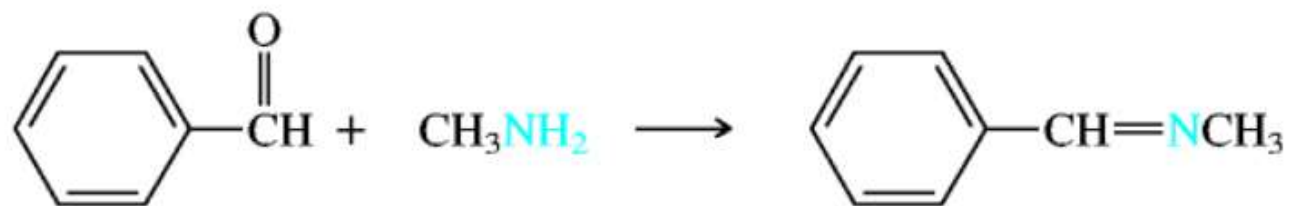
1. Imines and related compounds

1° amine

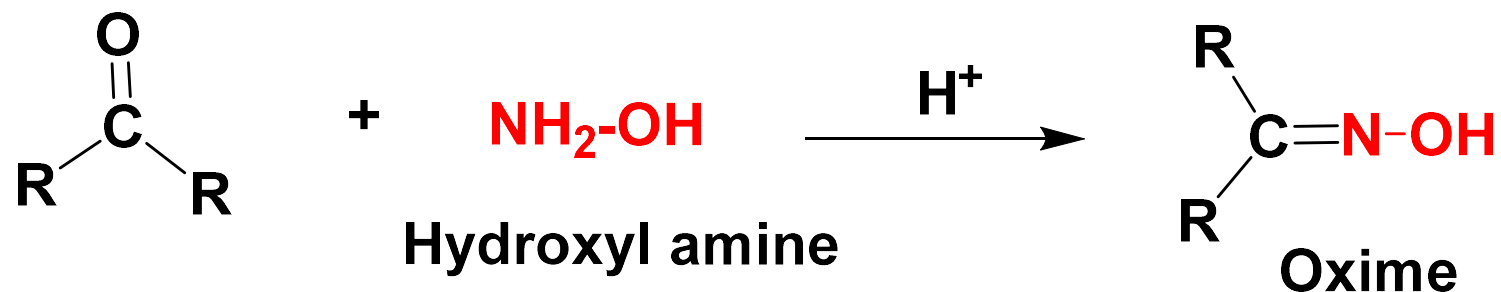


2° amine

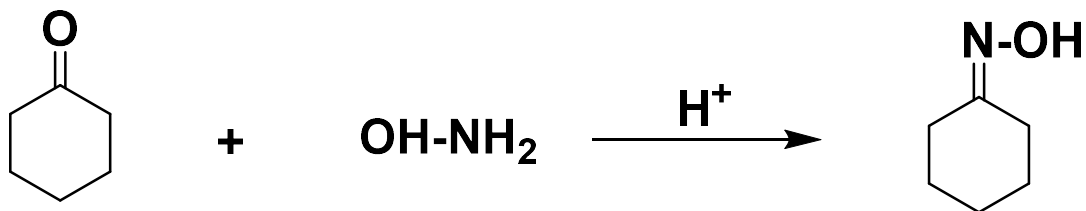
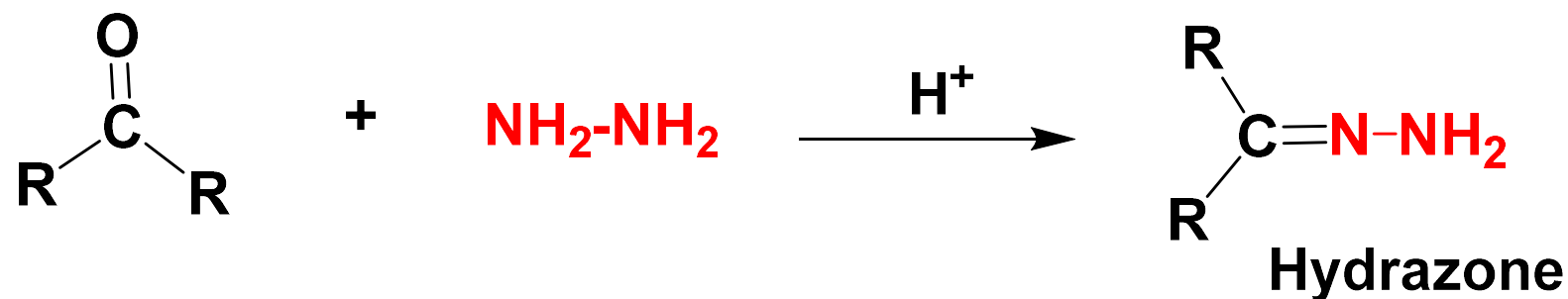




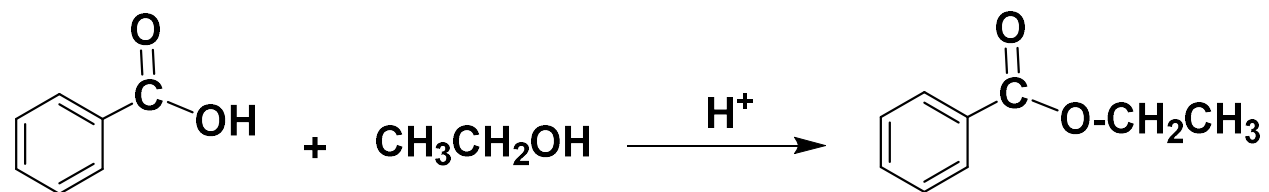
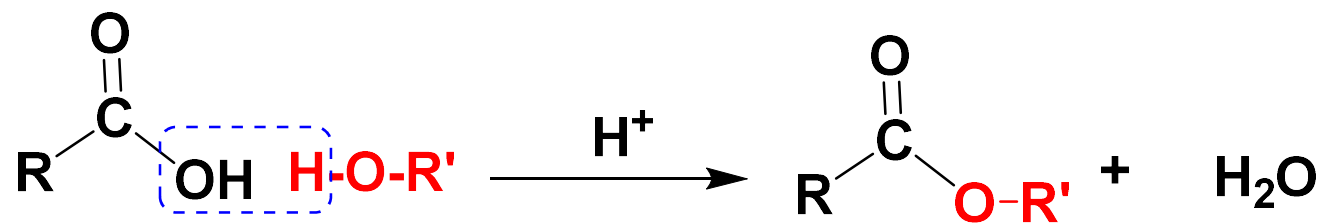
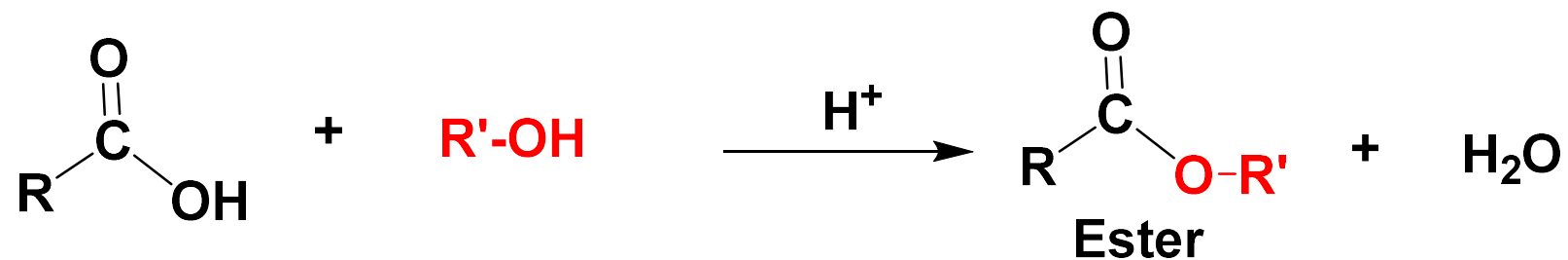
Oxime formation



Hydrazone formation

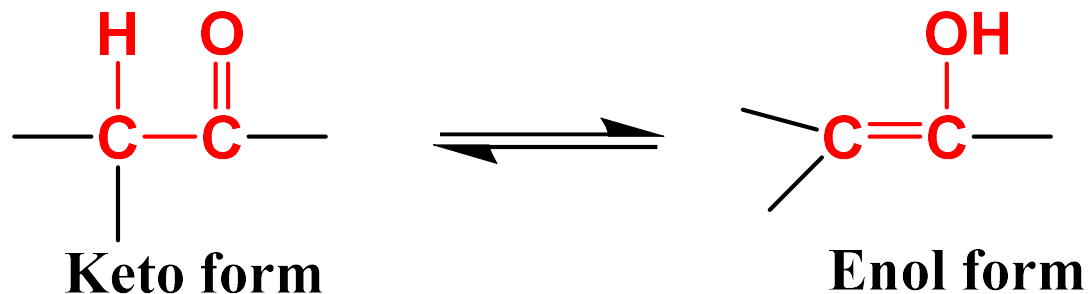


2. Ester Formation



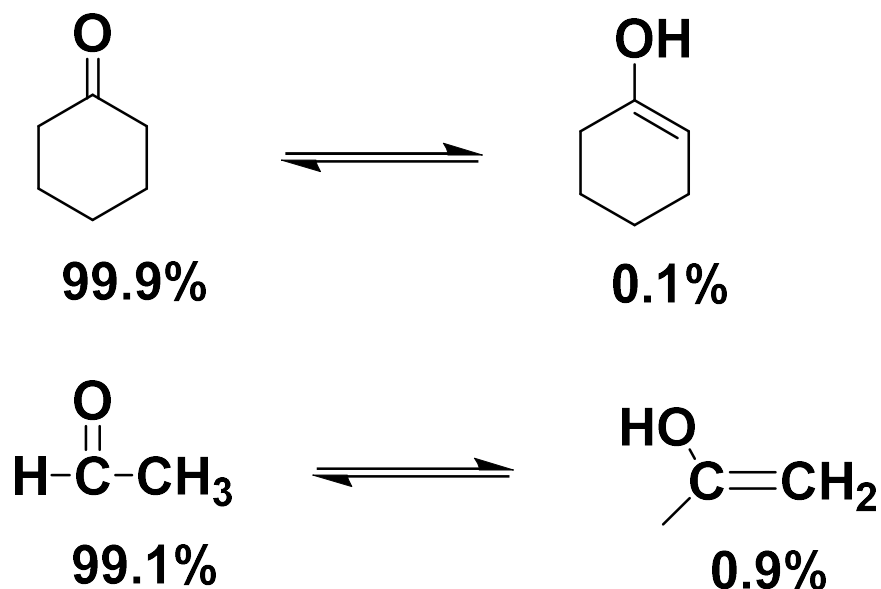
6.3 Enolization-ketolization reactions

Aldehydes and ketones may exist as an equilibrium mixture of two forms, called the **keto form** and the **enol form**.

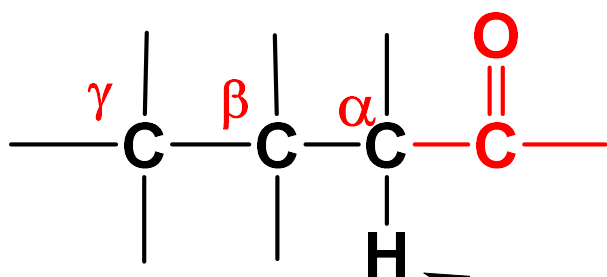


This type of structural isomerism is called **tautomerism**.

For simple ketones and aldehydes, the keto form predominates.

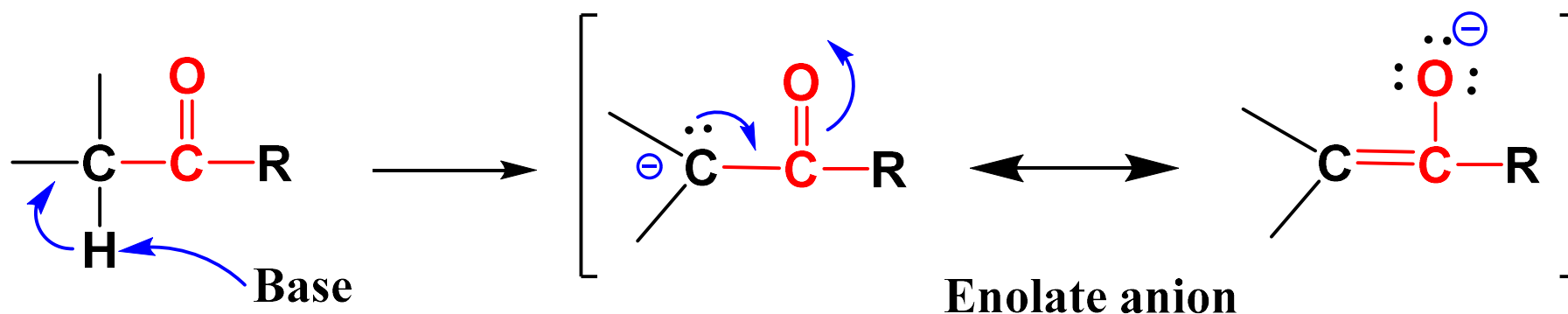


The α -hydrogen in a carbonyl compound is more acidic than a normal hydrogen bonded to a carbon atom.



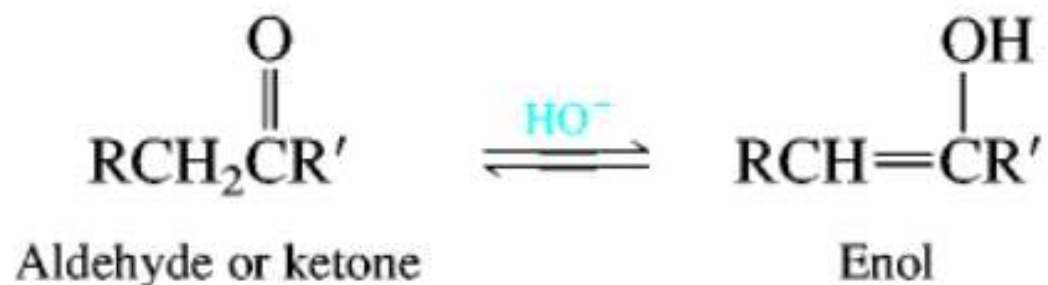
This proton is far more acidic than a hydrogen in an alkane.

This α -hydrogen can easily be removed by a base to form an enolate ion.

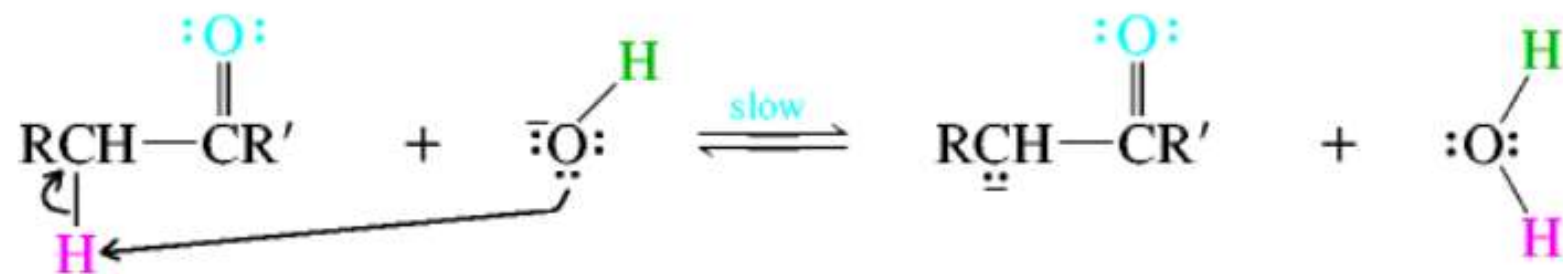


Enols are related to an aldehyde or a ketone by a proton-transfer equilibrium known as **keto-enol tautomerism**.

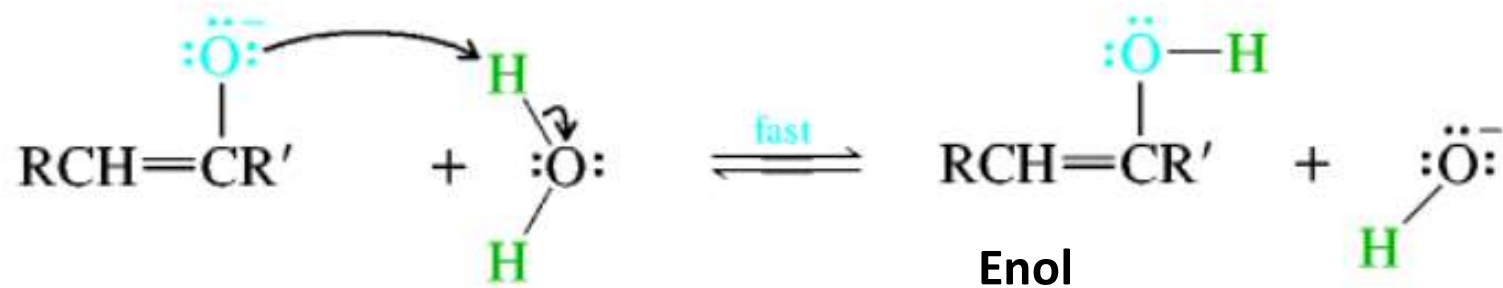
Base catalysed enolization



Step 1: A proton is abstracted by hydroxide ion from the α -carbon atom of the carbonyl compound.



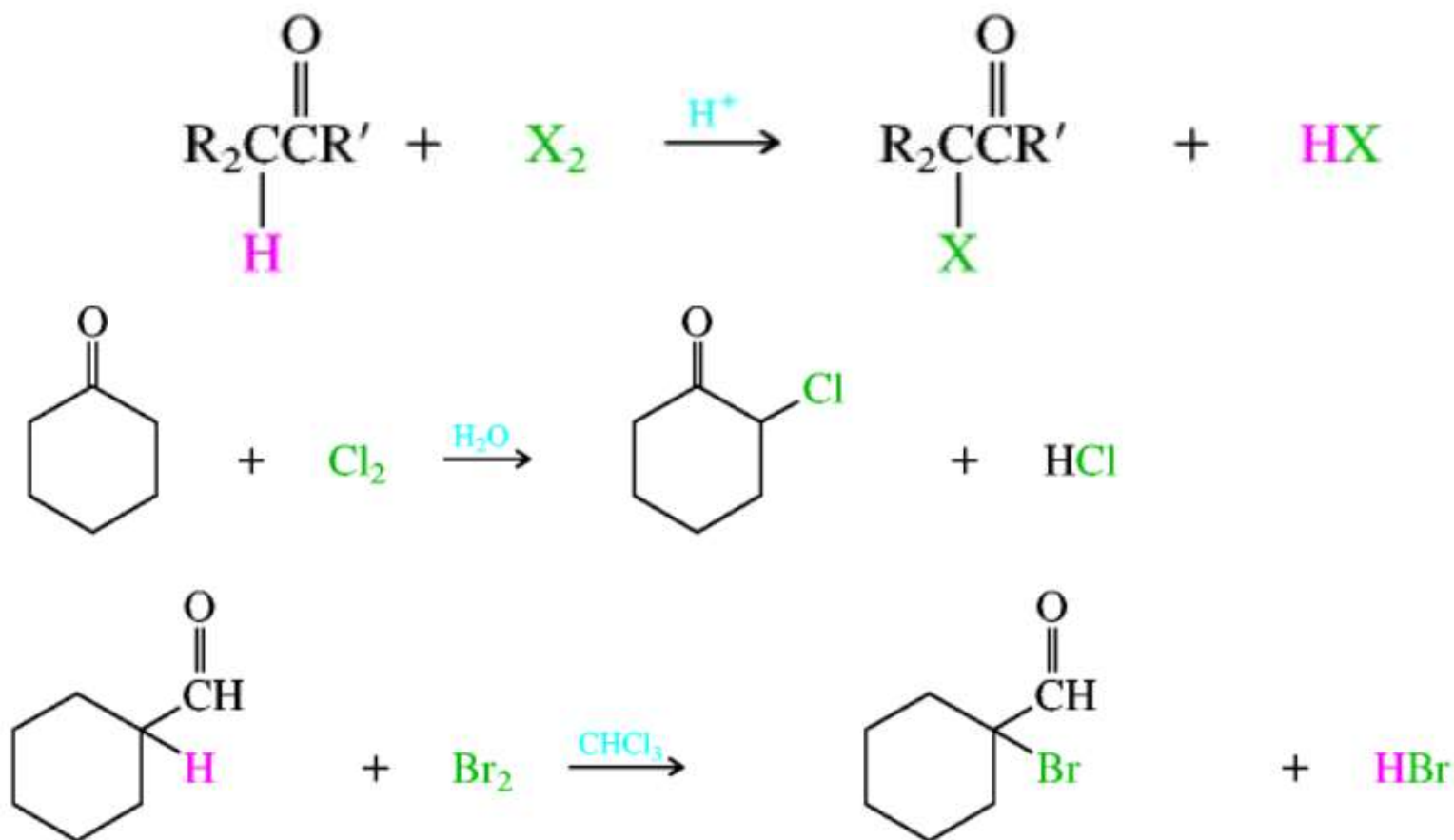
Step 2: A water molecule acts as a Brønsted acid to transfer a proton to the oxygen of the enolate ion.



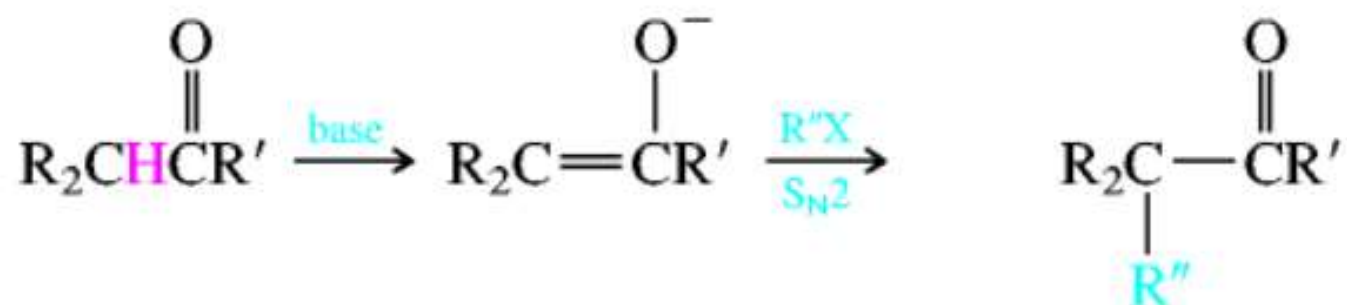
Application of enolization-Ketonization reactions

1. Halogenation

Aldehydes and ketones react with halogens by substitution of one of the α -hydrogens:

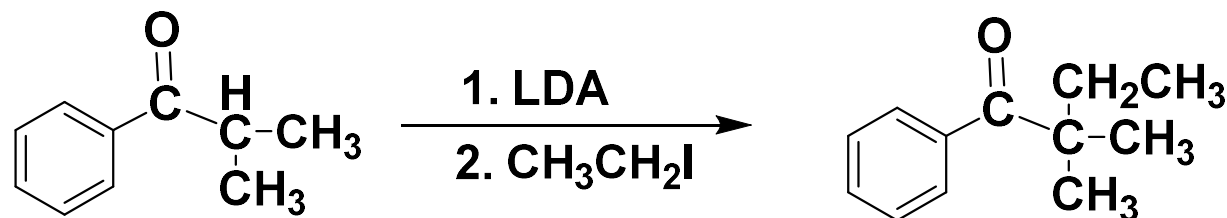
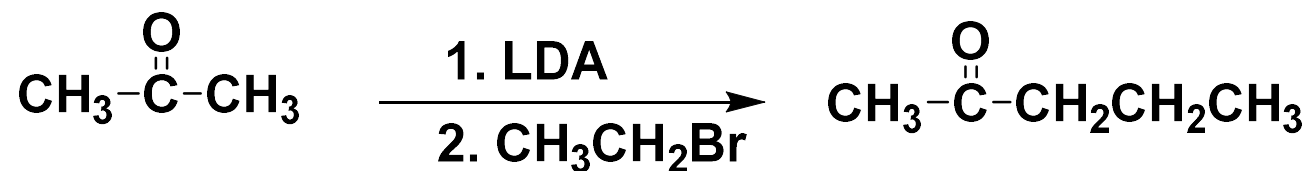


2. Alkylation of Enolate anions



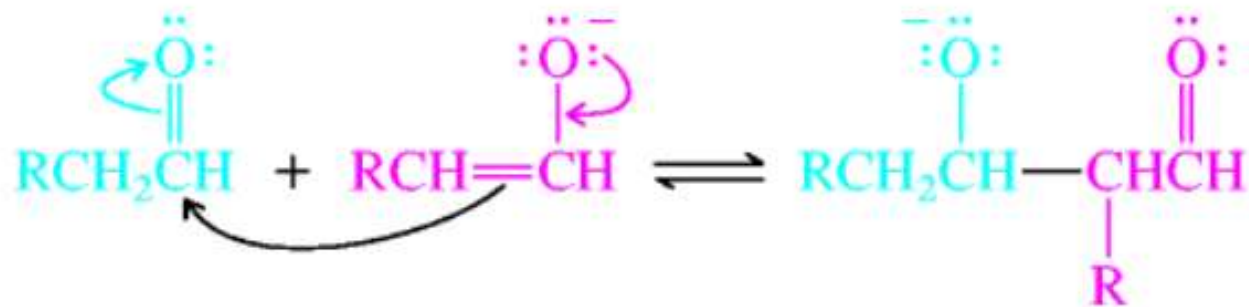
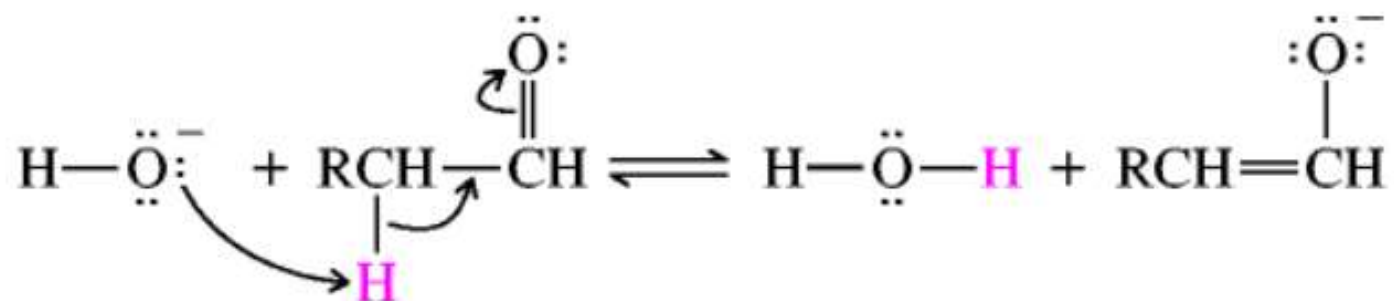
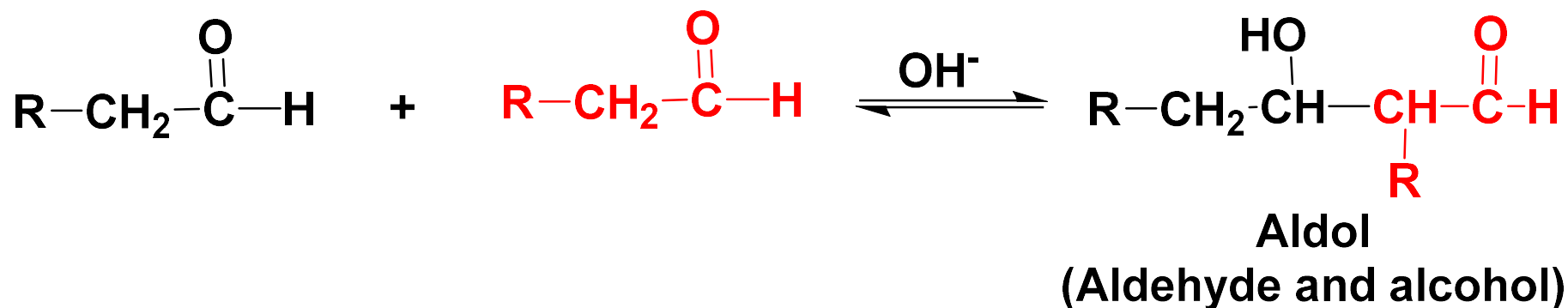
Base used for alkylation is LDA (Lithium diisopropyl amide)

$(\text{i-pr})_2\text{N}^- \text{Li}^+$ -bulky base
LDA -avoided side reaction

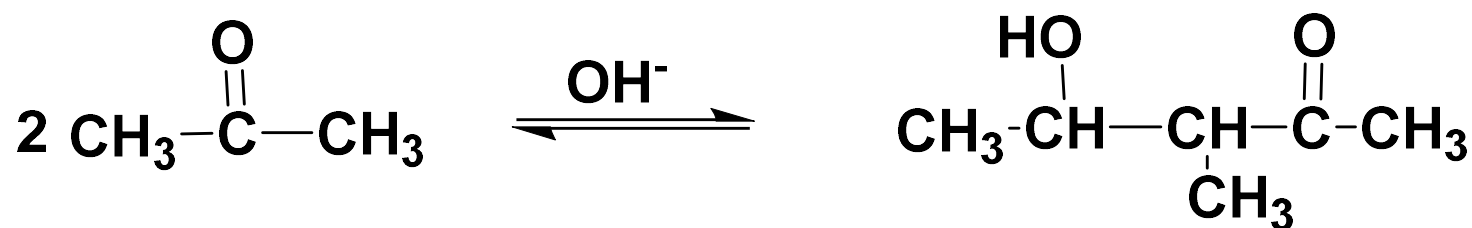
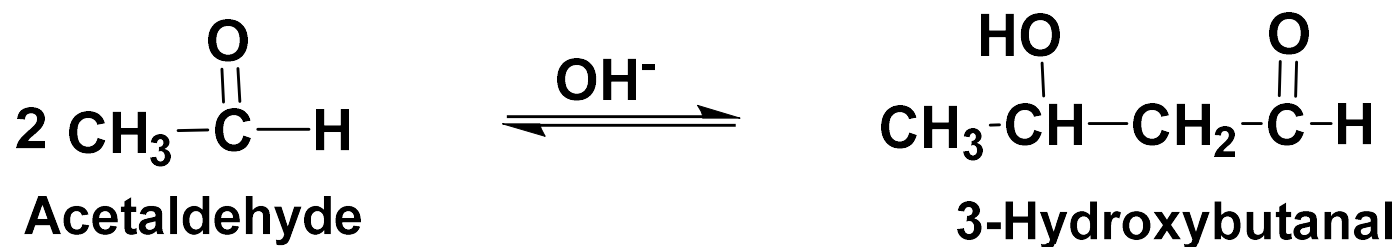
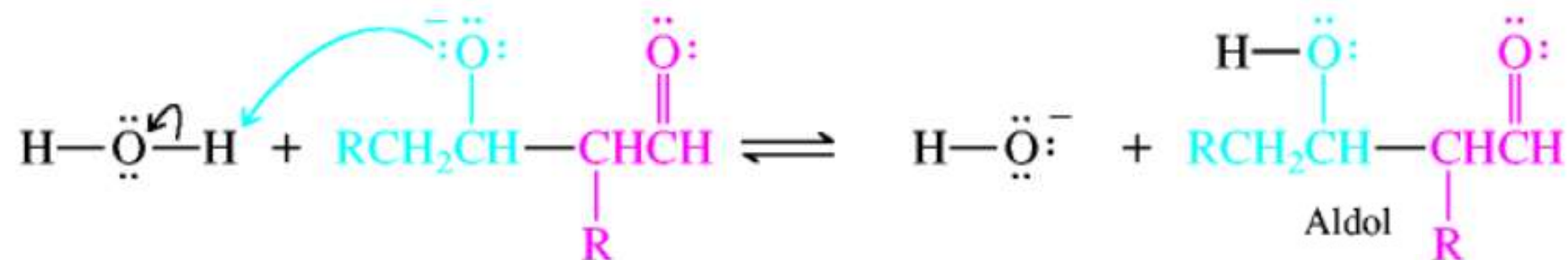


3. Aldol condensation

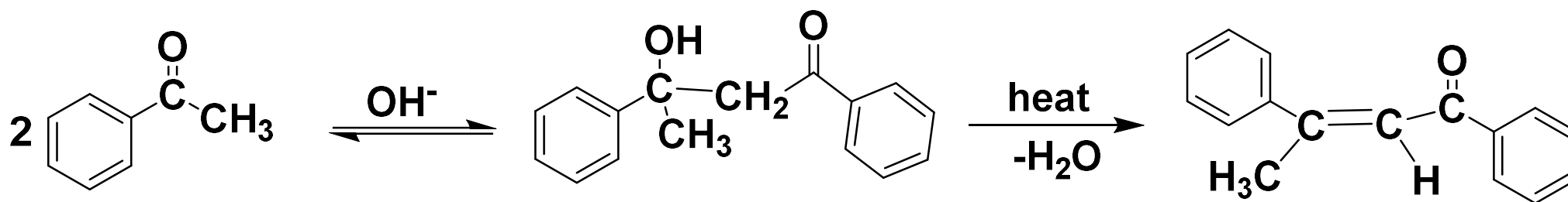
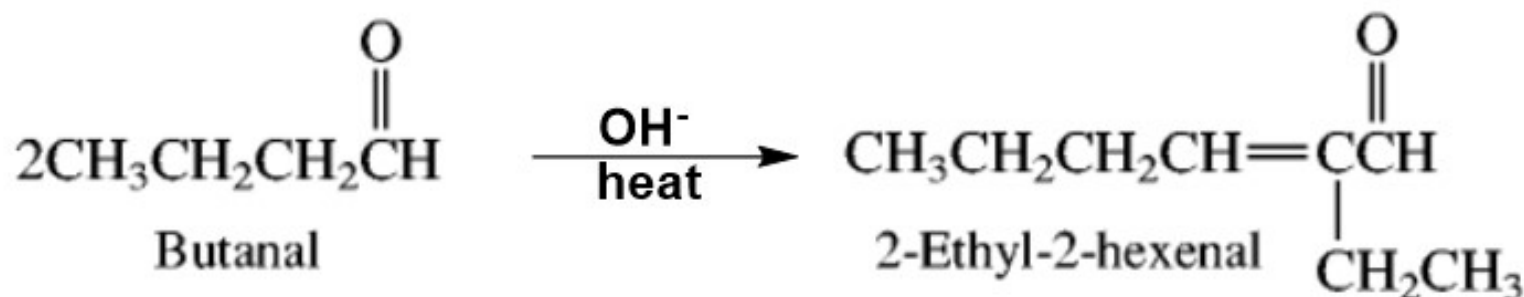
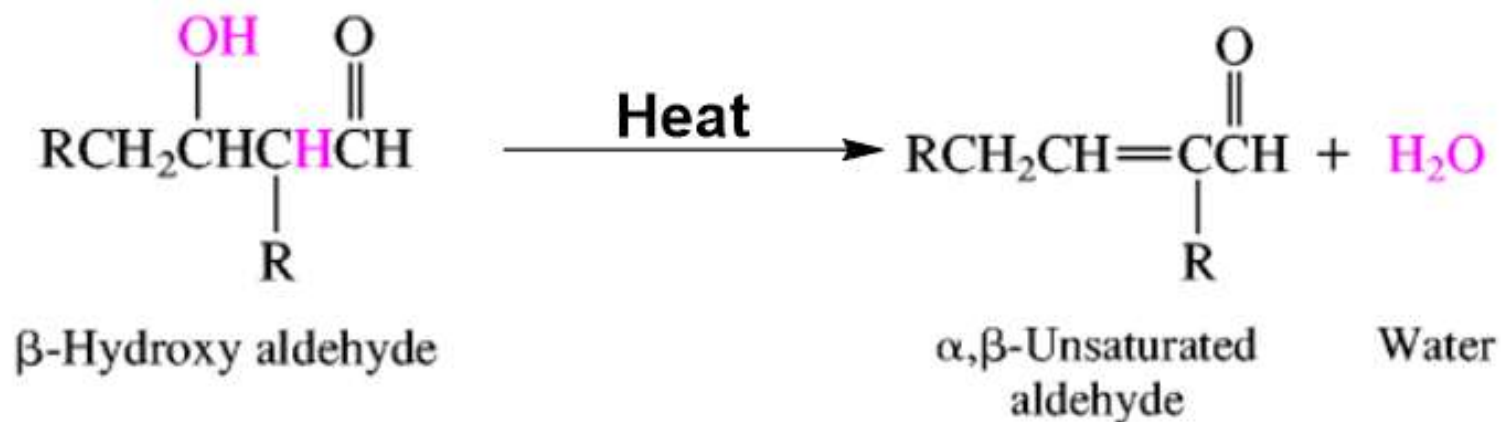
Is nucleophilic addition of an enolate ion in to another carbonyl.



The alkoxide ion abstracts a proton from water to give the product of aldol addition, a β -hydroxy aldehyde.



The β -hydroxy aldehyde products of aldol addition undergo dehydration on heating, to yield α, β -unsaturated aldehydes:

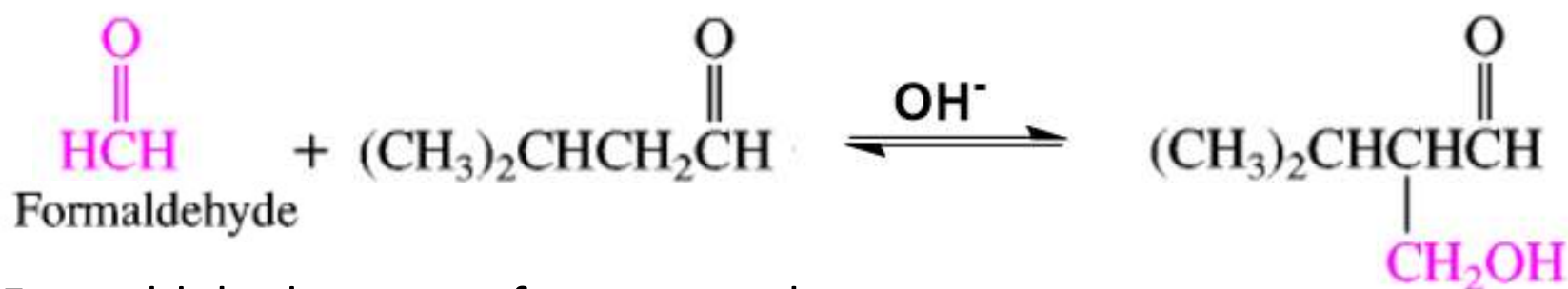


Mixed Aldol condensation

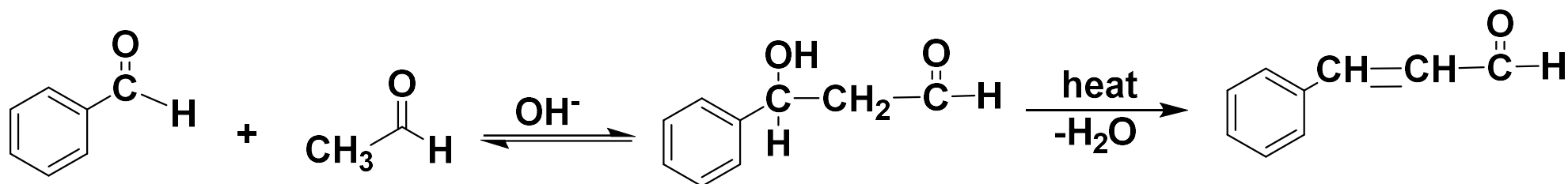
When different carbonyl reacts, mixture of aldol addition products possible.

The mixed aldol condensations that are the most synthetically useful are those in which:

1. Only one of the reactants can form an enolate; or
2. One of the reactants is more reactive toward nucleophilic addition than the other.

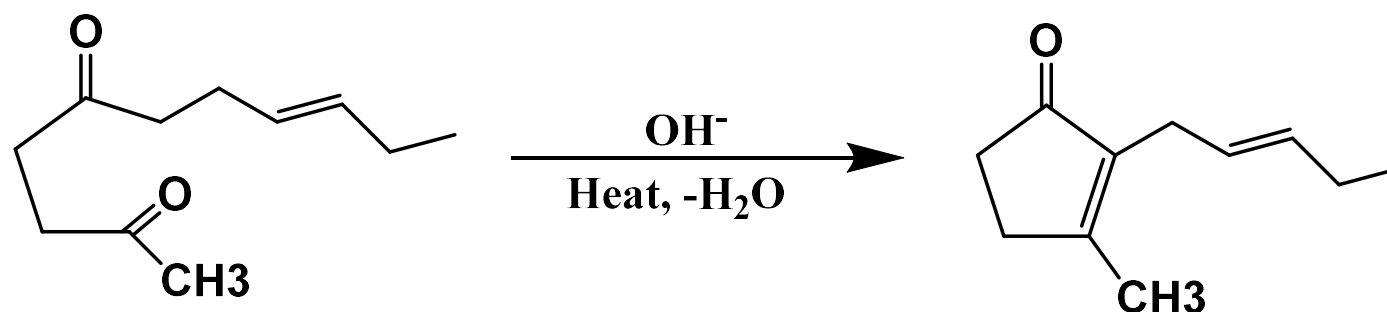
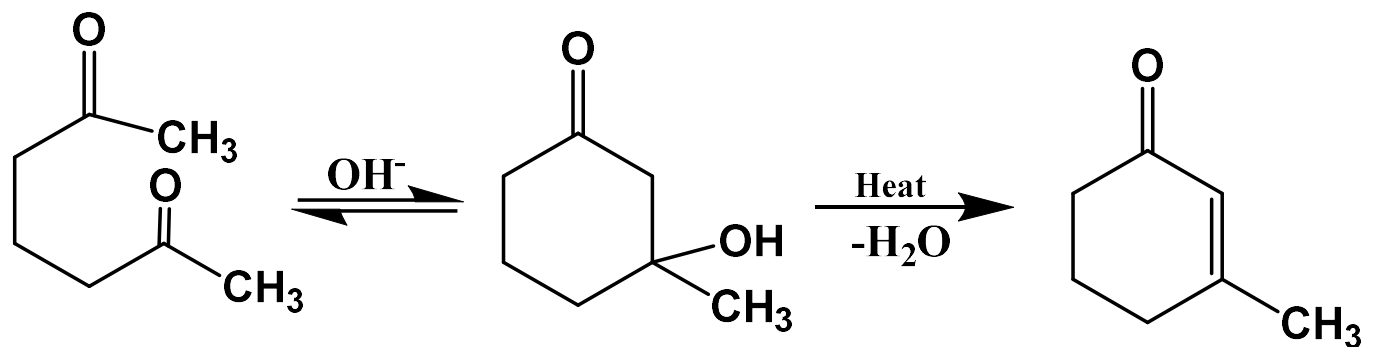


Formaldehyde cannot form an enolate



Aldol cyclization

- Intramolecular aldol reaction of diketones are useful for making 5 or 6-membered rings.
- Rings > 6 and < 5 are not common due to ring strain.

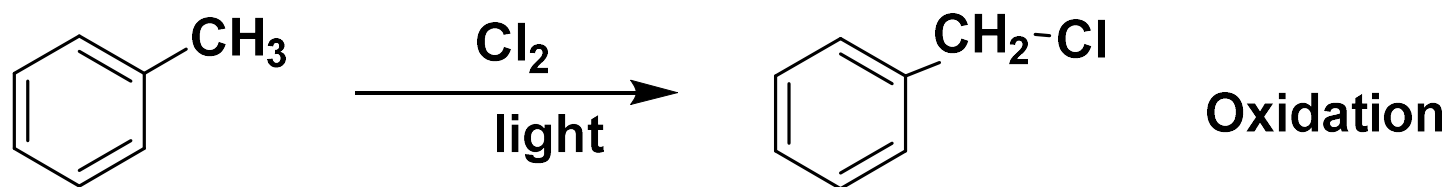
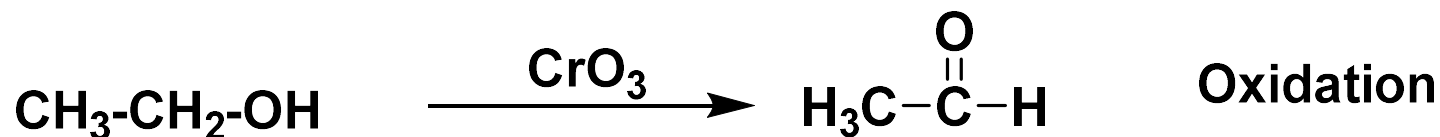


UNIT 7

OXIDATION-REDUCTION REACTION

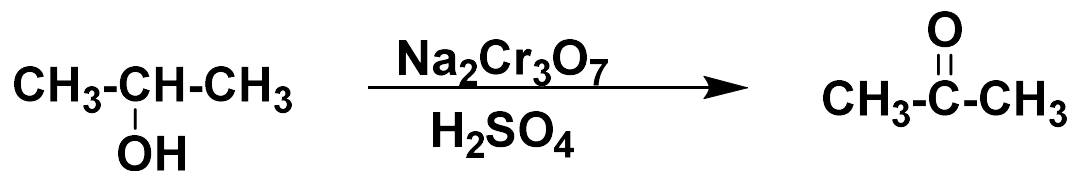
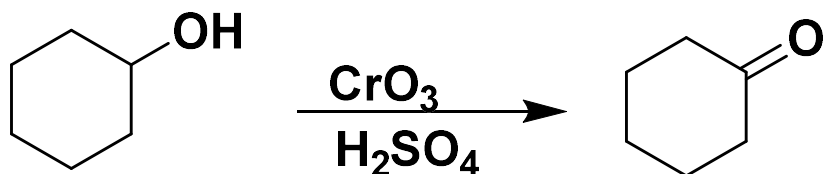
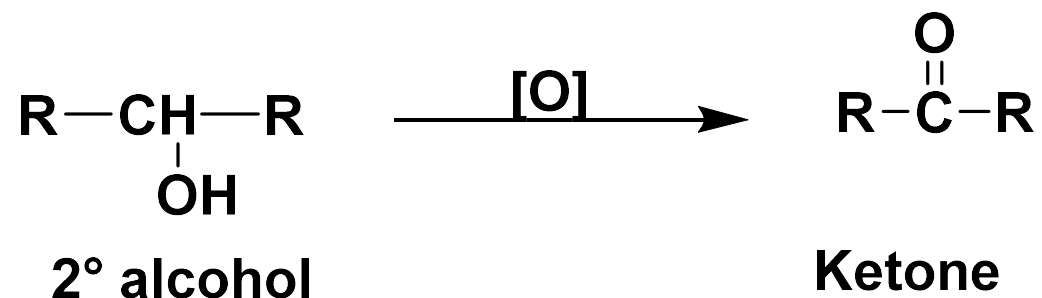
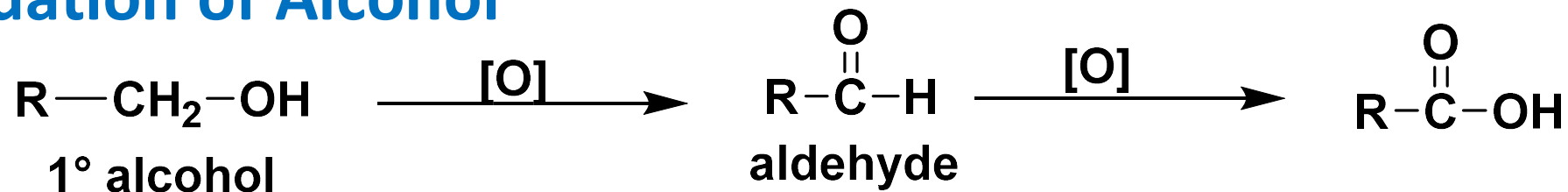
Oxidation reaction involves either gaining of oxygen atoms or loss of H-atoms.

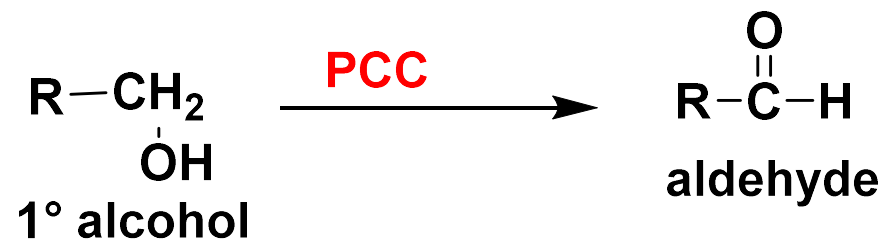
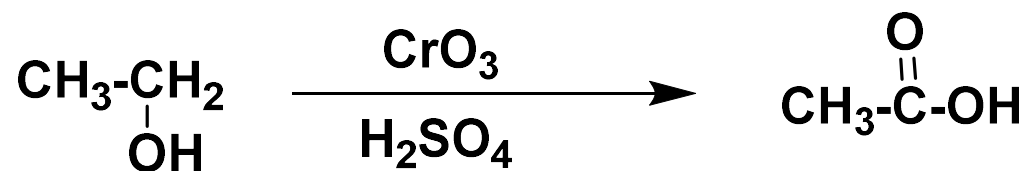
Reduction reactions involves gaining of H and losing of heteroatoms such as O, N, X.



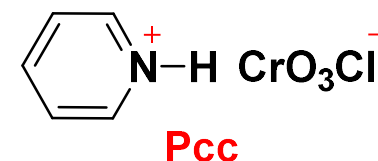
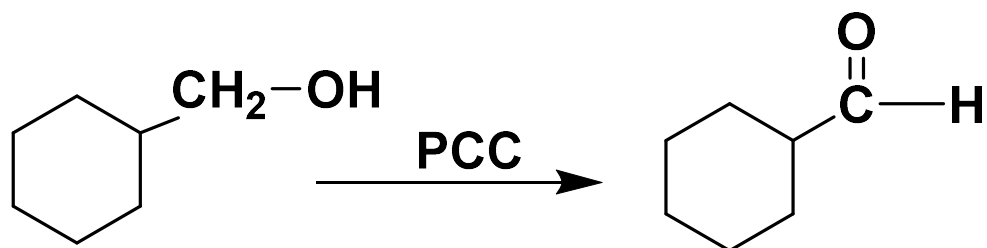
7.1 Oxidation Reaction

Oxidation of Alcohol





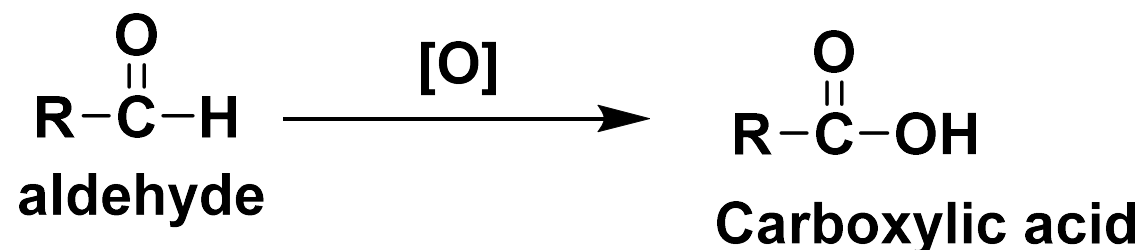
Pyridinium chlorochromate (PCC) stop oxidation at aldehyde



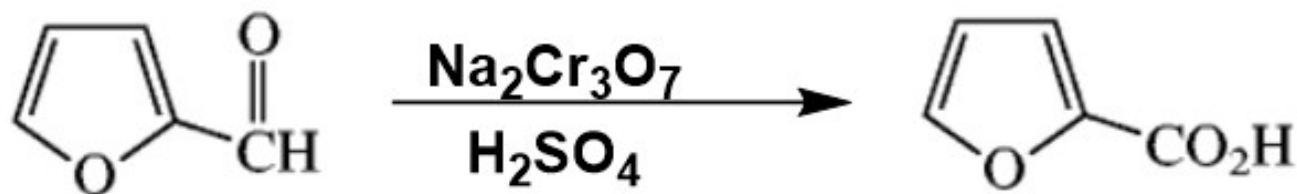
Biological Oxidation of Alcohols



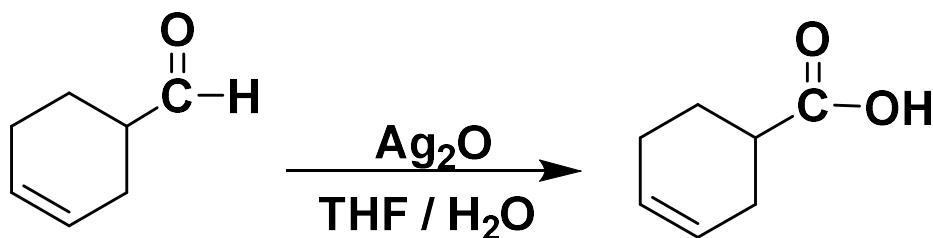
Oxidation of Aldehydes



$[\text{O}] = \text{CrO}_3, \text{KMnO}_4, \text{Na}_2\text{Cr}_2\text{O}_7, \text{RCOOH}$ (Peroxy acids)

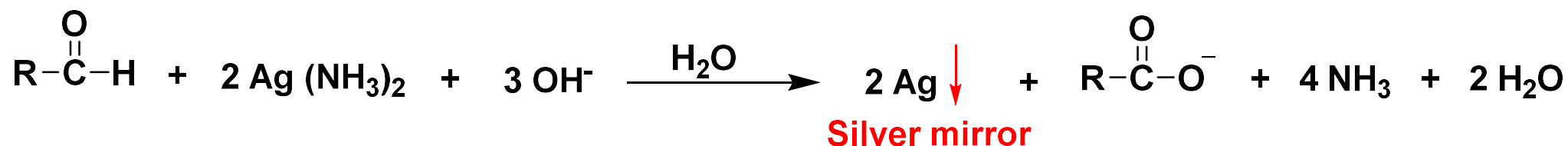


Mild oxidizing reagent like Ag_2O can selectively oxidize aldehyde in the presence of other functional group.



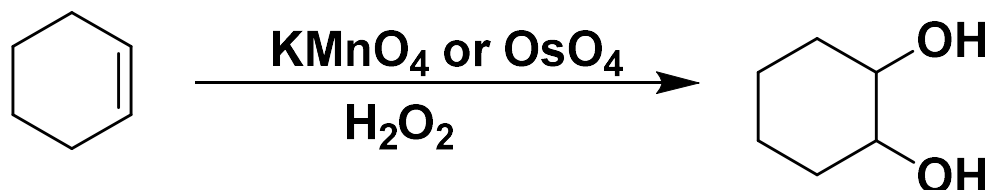
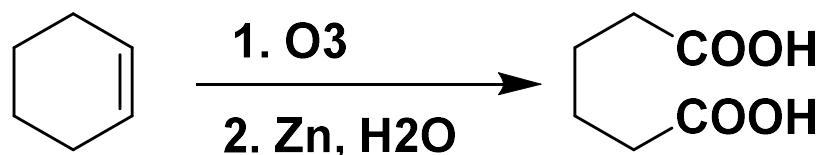
Tollen's Test

Test for aldehyde functional group



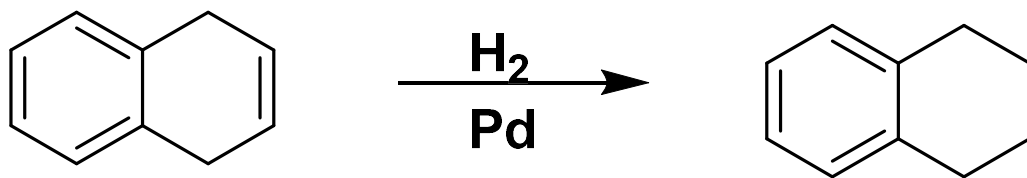
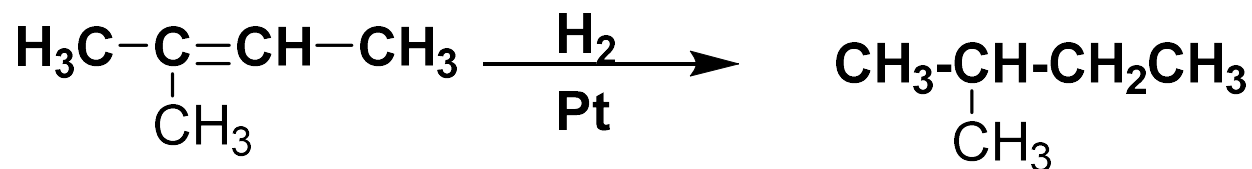
If a silver mirror is deposited on the wall of the container oxidation has occurred.

Oxidation of multiple bonds



7.2 Reduction Reaction

Catalytic hydrogenation



Hydride reduction (Chemical reduction)

